Venkata Lakshmi Parimala Pasupuleti

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Virginia, USA (open to NYC relocation)

PROFESSIONAL SUMMARY

Senior Data Engineer with 8+ years designing and owning large-scale data pipelines, ETL systems, and data infrastructure in housing technology. At Entrata, I owned end-to-end data pipelines powering eligibility, rent calculations, and compliance workflows for 100K+ monthly active users -- including the annual HUD income-limit ETL (few thousand federal rows expanded to ~180,000 domain records), RabbitMQ-based async processing with idempotency guarantees, and an AWS S3 document migration with 100% PHI compliance. I've optimized 200+ SQL queries (30% performance gain), designed schemas for 20+ production projects across multi-tenant environments, and built Python/Bash automation eliminating 80% of manual operational workload. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Large-Scale ETL Pipeline Design | Data Modeling & Database Design | Data Quality Monitoring & Checks | Client-Facing Reporting Solutions | Event-Driven Async Data Processing | AWS Data Infrastructure | Python & SQL at Scale | Cross-Functional Data Ownership

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building ML model pipelines and data analysis workflows; integrating data systems with back-end APIs; surfacing insights from raw business data into application-facing reporting layers.

George Mason University

Aug 2024 -- May 2026

Research Associate + Graduate Teaching Assistant

- Built multi-source prediction pipelines (Python, R) on WHO and UK Road Safety datasets across 190+ countries -- $R^2 = 0.977$ regression accuracy; designed full pipeline from data collection and transformation through validation.
- Applied Isolation Forest for automated data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards for faculty leadership (40% comprehension improvement).

Entrata (Housing Technology SaaS -- 100K+ monthly active users)

Jan 2017 -- Aug 2024

Senior Software Engineer (Data & Backend) -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Owned the core ETL pipeline for housing operations data end-to-end for 7 years:** annual HUD income-limit import expanding few thousand federal rows into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units. Every release included baseline validation, reconciliation, and rollback preparation -- one data error affects thousands of residents' eligibility decisions. Zero compliance violations across the ownership period.
- **Designed and maintained large-scale data pipelines and data models** for leasing, financial, compliance, and document workflows: RabbitMQ-based async ETL with idempotency and retry guarantees; schema design for 20+ production projects across multi-tenant PostgreSQL environments; 200+ SQL query optimizations (30% performance gain) via execution plan analysis, indexing, partitioning, and stored procedure refactoring.
- **Built data quality checks and monitoring systems:** internal debugging tooling that cut mean debug time 40%; Python/PL-SQL automation eliminating 80% of manual operational workload; maintained bug count below 10 for 6+ consecutive months. Implemented AWS S3 document migration (100% PHI compliance, zero security incidents) with full data lineage and validation.
- **Developed client-facing reporting solutions:** 56 state-specific housing certification documents with merge fields; financial reconciliation reports for affordable housing compliance; cross-functional dashboards surfacing eligibility and portfolio data for operations teams and customers.
- Mentored 5+ engineers; conducted 300+ peer code reviews; collaborated across product, operations, and analytics teams on data-driven features deployed across 120+ Agile sprints.

PROJECTS

Leasel: AI-Powered Data Intelligence Platform for Housing Operations

LangGraph -- FastAPI -- RAG -- SQLite -- Python

Built an AI platform that extracts structured data from unstructured housing documents, runs compliance validation pipelines, and surfaces evidence-backed hypotheses from historical data. Architecture: hybrid TF-IDF + semantic RAG with citation grounding; SQLite knowledge repository; 8-agent LangGraph orchestration with HITL escalation; data quality guardrails blocking unsupported outputs. Results: 93.6% precision / 87.9% F1 on 1,240 test cases across 5 US states; 100% citation correctness.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: AI Pipeline Evaluation & Data Quality Toolkit

FastAPI -- Python -- ChromaDB -- CI/CD

15-endpoint FastAPI backend with a 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate with golden datasets, and multi-provider LLM router. Built to measure whether data-driven AI pipelines are producing correct, grounded outputs -- used to validate Leasel across 1,240+ test cases. Zero vendor dependency, fully local.

github.com/venkatalakshmiparimala/AgentProbe

CrashClarity: Large-Scale Road Accident Data Pipeline (104K Records) Python -- R -- scikit-learn -- Feature Engineering

End-to-end data pipeline on 104K+ UK accident records -- data collection, cleaning, feature engineering, Random Forest + Logistic Regression modeling (85%+ accuracy), and interactive dashboard deployed on GitHub Pages.

venkatalakshmiparimala.github.io/Portfolio/project_crashclarity.html

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

Data Engineering: ETL Design & Implementation, Data Pipeline Architecture, Data Modeling, Dimensional Modeling, Medallion Architecture, Snowflake, PostgreSQL, MySQL, Oracle, T-SQL, PL/SQL, Apache Airflow, Apache Kafka, RabbitMQ, Data Quality Frameworks, Data Lineage, Schema Migrations

Languages & Frameworks: Python, SQL, R, TypeScript, JavaScript, FastAPI, Django, Spring Boot, Bash

Cloud & Infrastructure: AWS (S3, Glue, EC2, RDS, Lambda, Step Functions), Azure Data Factory, Azure Data Lake, Docker, Kubernetes, CI/CD (GitHub Actions), Terraform, PostgreSQL, Redis, Linux

Analytics & Visualization: Tableau, Power BI, ggplot2, Plotly, Pandas, NumPy, scikit-learn, PyTorch, Jupyter, Statistical Modeling, Feature Engineering